What is the Django Template Engine?

Django Template Engine: The Django Template Engine is a system that helps developers create dynamic HTML pages. It allows us to display data from Python views inside HTML templates.

Like <h1>My name is {{name}}</h1>

Template Tags: Template tags are used to perform logical operations inside HTML templates, such as conditions and loops.

```
Like:  {% if age>=18%}
           <h1>You are adult</h1>
       {%else%}
          <h1>You are a baby</h1>
       {%endif%}
```

Template Filters: Template filters are used to modify or format the value of a variable before displaying it.

Like <h1>My name is {{name | upper}}</h1>

What is Django Middleware?

Django Middleware: Django Middleware is a component that works between the web server and Django's view system. It processes incoming requests before they reach the view and processes responses before they are sent back to the browser.

Built-in Middleware: Built-in Middleware is provided by Django itself. Developers can enable it through the MIDDLEWARE list in settings.py.

Like: SecurityMiddleware , SessionMiddleware , MessageMiddleware

Custom Middleware: Custom Middleware is created by developers to perform specific tasks according to project requirements.

Middleware Execution Order:

1. When a request comes from the browser, middleware processes it from top to bottom.
2. After passing through all middleware, the request reaches the view.
3. The view processes the request and generates a response.
4. The response passes through middleware in reverse order, from bottom to top.
5. Finally, the response is returned to the browser.

What is a Django ModelForm?

Django ModelForm : A ModelForm is a Django class that automatically creates an HTML form using an existing Django model.

Creating a ModelForm:

```
Class Student(models.Model):
    Name = models.CharField(max_length=50)
    Email = models.EmailField(unique=True)
    Age = models.IntegerField()
```

Create a ModelForm:

```
class StudentForm(forms.ModelForm):

    class Meta:

        model = student

        fields = '__all__'
```

Handle Form Submission:

```
def student_view(request):

    If request.method == "POST":

        form = StudentForm(request.POST)

        If form.is_valid():

            form.save()
```

```
                return redirect('student')
        else:
                form = StudentForm()
```

Create an HTML Template:

```
<form method='post'>
        {%csrf_token%}
        {{form.studentform}}
        <button type='submit'>Submit</button>
```

What is the difference between Static Files and Media Files in Django?

Static Files: Static files are files used to design and improve the appearance of a website. These files are usually created and managed by developers.

Like : CSS files , JavaScript files etc

Media Files: Media files are files uploaded by users through the website.

Like : profile picture , product image etc

Configuring Static Files:

```
STATIC_URL = 'static/'
STATICFILES_DIRS = [
        BASE_DIR / 'static'
]
```

Configuring Media Files:

```
MEDIA_URL = '/media/'
MEDIA_ROOT = BASE_DIR / 'media'
```

Explain Django Authentication and Session Middleware. How do Login, Logout, and Session Management work in Django?

Django Authentication: Django Authentication is a built-in system that helps manage users and control access to a website.

Providing features:
1.user registration
2.user login
3.user logout
4.password management

Session Middleware: Session Middleware is responsible for managing user sessions in Django.It allows Django to remember information about a user across multiple requests.

Login in Django: Django provides a built in login() function to authenticate and log in users.

Logout in Django: Django provides a built-in logout() function to log out users and clear their authenticated session.

Session Management: Session management allows Django to store and retrieve information related to a particular user between requests.